ACCENTURE STRATEGY BRIEF COMPETITION PROMPT

PROJECT REFLOW: DIGITAL SUPPLY CHAIN TRANSFORMATION

Executive Case Competition Problem Statement & Operational Dataset

CASE OVERVIEW: NorthStar Retail Group is a major Canadian retail enterprise generating CAD $1.2 Billion in annual revenue across department stores, grocery, and e-commerce. Fulfillment is managed through 3 regional Distribution Centres (DCs) shipping 12.0 million orders annually. Over the past 24 months, surging parcel volumes have overwhelmed legacy manual processes, causing an 8.5% late delivery rate, $86.4M in annual logistics spend ($7.20/order), and 18,000 tonnes of Scope 1 greenhouse gas emissions. The Board of Directors has established a CAD $20.0M capital allocation ceiling and mandates a 3-year simple payback horizon.

CAD $1.2B
ANNUAL REVENUE
3 Regional Divisions

12.0M Orders
ANNUAL FULFILLMENT
40,000 Orders Daily

CAD $20.0M
CAPEX CEILING
Board Authorized Budget

<= 3.0 Yrs
PAYBACK MANDATE
Strict Capital Hurdle

SECTION 1: THE THREE CANDIDATE INITIATIVES

OPTION A: AUTOMATED DC PICKING (ROBOTICS)

Upfront Capex: CAD $8.0M | Recurring OpEx: CAD $0.5M/yr | Gross Savings: CAD $5.4M/yr

Deploys vision-guided mobile robots (cobots) in DC 1 and DC 2 to eliminate 38% picker walk travel drag. Reduces late delivery rate by 2.0 percentage points and cuts logistics CO2e by 8% through facility energy optimization. Net Annual Run-Rate: CAD $4.90M. Simple Payback: 19.6 Months.

OPTION B: INTELLIGENT ROUTING & FLEET TELEMETRY (ML)

Upfront Capex: CAD $5.0M | Recurring OpEx: CAD $0.3M/yr | Gross Savings: CAD $4.2M/yr

Integrates carrier APIs and dynamic route clustering algorithms. Increases trailer cube utilization from 62% to 84%, eliminating empty haul miles. Reduces late delivery rate by 3.2 percentage points and cuts fleet diesel emissions by 18.0%. Net Annual Run-Rate: CAD $3.90M. Simple Payback: 15.4 Months.

OPTION C: REGIONAL MICRO-FULFILLMENT HUBS

Upfront Capex: CAD $12.0M | Recurring OpEx: CAD $2.2M/yr | Gross Savings: CAD $6.0M/yr

Constructs 5 leased urban micro-fulfillment depots closer to core metropolitan populations. Compresses final-mile stem mileage, reducing late delivery rate by 3.5 percentage points and carbon by 12.0%. However, high ongoing lease/labor OpEx ($2.2M) extends simple payback to 37.9 Months (breaching the 3-year mandate). Net Annual: CAD $3.80M.

SECTION 2: MANDATORY CASE CHALLENGE QUESTIONS

1. **Question 1: Portfolio Selection & Capital Allocation**
   Which combination of Options A, B, and C should NorthStar execute to maximize net financial value without exceeding the $20.0M capex ceiling? Justify with exhaustive ranking.
2. **Question 2: Deterministic 3-Year DCF Proforma**
   Construct a complete 3-year cash flow model using a 10.0% WACC discount rate. Define NPV, simple payback period, and net annual operating run-rates.
3. **Question 3: Operational Decarbonization Physics**
   Demonstrate how the selected strategy reduces NorthStar's logistics carbon footprint by at least 20.0% and cuts late delivery below 4.0%.
4. **Question 4: Phased Execution Roadmap & Risk Matrix**
   Structure a 12-month implementation timeline across 3 Horizons with predictive leading and financial lagging indicators. Include a 2x2 risk pre-mortem.

THE OFFICIAL WINNING STRATEGY: PORTFOLIO A+B

Complete evaluation proving why Portfolio A+B dominates all alternate configurations

SEVEN CANDIDATE PORTFOLIOS EVALUATION SUMMARY

Portfolio A+B (Champion)	**CAD $13.0M**	**CAD $8.08M/yr**	**19.3 Mo**	**+$7.094M**	3.5% Late	**APPROVED**
Option B (Solo)	CAD $5.0M	CAD $3.90M/yr	15.4 Mo	+$4.698M	5.3% Late	**Viable (Small)**
Option A (Solo)	CAD $8.0M	CAD $4.90M/yr	19.6 Mo	+$4.185M	6.5% Late	**Viable**
Portfolio B+C	CAD $17.0M	CAD $7.45M/yr	27.4 Mo	+$1.528M	3.5% Late	**Sub-Optimal**
Portfolio A+C	CAD $20.0M	CAD $8.40M/yr	28.6 Mo	+$0.899M	3.5% Late	**Capital Heavy**
Option C (Solo)	CAD $12.0M	CAD $3.80M/yr	37.9 Mo	-$2.548M	5.0% Late	**Fails Payback**
Portfolio A+B+C	CAD $25.0M	CAD $11.8M/yr	25.4 Mo	N/A	3.5% Late	**Breaches Cap**

WHY PORTFOLIO A+B IS THE DETERMINISTIC WINNER

1. Superior Net Present Value (+CAD $7.094M at 10% WACC):

Portfolio A+B generates 4.6x the NPV of Portfolio B+C ($1.528M) and 7.9x that of Portfolio A+C ($0.899M), while Option C solo destroys value (-$2.548M).

2. Preserves CAD $7.00M in Capital Contingency Liquidity:

Commitment of $13.0M leaves 35% of the authorized $20.0M budget untouched in cash reserves, completely eliminating corporate debt or equity dilution risks.

3. Compresses Late Deliveries to 3.5% (Meeting -5.0 pt Requirement):

Combined reductions from warehouse picking (-2.0 pts) and transportation routing (-3.2 pts) total -5.2 pts, capped by law at the 5.0 pt threshold, achieving a 3.5% late rate.

4. Exceeds ESG Carbon Goals (-25.0% Fleet CO2e Decarbonization):

Saves 4,500 tonnes of Scope 1 greenhouse gas emissions annually, far surpassing corporate sustainability mandates and creating $225,000 in carbon offset credit value.

5. Simple Payback in 19.3 Months (1.61 Years Horizon):

Recovers the full $13.0M capital investment nearly 17 months before the board's 3-year deadline, generating $8.08M in perpetual recurring annual pre-tax savings.